

Bank Loan Exploratory Data Analysis

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Loading the data

Lets start by loading our dataset with the readxl package.

```
library(readxl)
dataset <- data.frame(read_xlsx("bank_loan_default.xlsx"))
```

Initial Data Check

In this initial data check, we take a quick look at the structure and key characteristics of our dataset. First, we preview the first and last few rows using head and tail to get a feel for how the data is organized. Then, we check the dataset's dimensions with dim, which reveals it has 1,000 observations and 16 variables. The str function gives an overview of each variable's data type and a few sample values, helping us differentiate between numeric and categorical fields. Finally, summary provides a concise statistical snapshot, showing measures like the minimum, maximum, mean, and quartiles for numeric variables, while summarizing lengths for categorical ones. Together, these steps help us understand the dataset's layout, detect any unusual values, and identify potential areas for data cleaning before deeper analysis.

```
head(dataset)
```

##	Default	Checking_amount	Term	Credit_score	Gender	Marital_status	Car_loan
## 1	0	988	15	796	Female	Single	1
## 2	0	458	15	813	Female	Single	1
## 3	0	158	14	756	Female	Single	0
## 4	1	300	25	737	Female	Single	0
## 5	1	63	24	662	Female	Single	0
## 6	0	1071	20	828	Male	Married	1

##	Personal_loan	Home_loan	Education_loan	Emp_status	Amount	Saving_amount
## 1	0	0	0	employed	1536	3455
## 2	0	0	0	employed	947	3600
## 3	1	0	0	employed	1678	3093
## 4	0	0	1	employed	1804	2449
## 5	0	0	1	unemployed	1184	2867
## 6	0	0	0	employed	475	3282

##	Emp_duration	Age	No_of_credit_acc
## 1	12	38	1
## 2	25	36	1
## 3	43	34	1
## 4	0	29	1

```
## 5          4 30          1
## 6          12 32          2
```

```
tail(dataset)
```

```
##      Default Checking_amount Term Credit_score Gender Marital_status
Car_loan
## 995      0          589    20          733    Male      Married
1
## 996      1          17    21          562    Male      Married
1
## 997      0          590    18          873 Female      Single
1
## 998      0          343    16          824    Male      Married
0
## 999      0          709    16          811    Male      Married
1
## 1000     0          991    15          794    Male      Married
0
```

```
##      Personal_loan Home_loan Education_loan Emp_status Amount
Saving_amount
## 995          0          0          0 unemployed    829
3171
## 996          0          0          0 unemployed    1328
2658
## 997          0          0          0 unemployed    1433
3469
## 998          1          0          0 unemployed    1465
3426
## 999          0          0          0 unemployed    1359
3114
## 1000         1          0          0 unemployed    1321
3309
```

```
##      Emp_duration Age No_of_credit_acc
## 995          70 33          4
## 996         118 28          5
## 997         108 29          5
## 998          99 31          4
## 999         113 28          6
## 1000         95 33          8
```

```
dim(dataset)
```

```
## [1] 1000  16
```

```
str(dataset)
```

```
## 'data.frame':  1000 obs. of  16 variables:
## $ Default      : num  0 0 0 1 1 0 0 0 0 1 ...
## $ Checking_amount : num  988 458 158 300 63 ...
## $ Term         : num  15 15 14 25 24 20 13 16 20 19 ...
```

```
## $ Credit_score      : num  796 813 756 737 662 828 856 763 778 649 ...
## $ Gender            : chr   "Female" "Female" "Female" "Female" ...
## $ Marital_status    : chr   "Single" "Single" "Single" "Single" ...
## $ Car_loan          : num    1 1 0 0 1 1 1 1 1 ...
## $ Personal_loan     : num    0 0 1 0 0 0 0 0 0 ...
## $ Home_loan         : num    0 0 0 0 0 0 0 0 0 ...
## $ Education_loan    : num    0 0 0 1 1 0 0 0 0 ...
## $ Emp_status        : chr   "employed" "employed" "employed" "employed" ...
## $ Amount            : num  1536 947 1678 1804 1184 ...
## $ Saving_amount     : num  3455 3600 3093 2449 2867 ...
## $ Emp_duration      : num   12 25 43 0 4 12 11 12 12 0 ...
## $ Age               : num   38 36 34 29 30 32 38 36 36 29 ...
## $ No_of_credit_acc  : num    1 1 1 1 1 2 1 1 1 1 ...
```

summary(dataset)

```
##      Default      Checking_amount      Term      Credit_score
## Min.      :0.0      Min.      :-665.0      Min.      : 9.00      Min.      : 376.0
## 1st Qu.:0.0      1st Qu.: 164.8      1st Qu.:16.00      1st Qu.: 725.8
## Median :0.0      Median : 351.5      Median :18.00      Median : 770.5
## Mean      :0.3      Mean      : 362.4      Mean      :17.82      Mean      : 760.5
## 3rd Qu.:1.0      3rd Qu.: 553.5      3rd Qu.:20.00      3rd Qu.: 812.0
## Max.      :1.0      Max.      :1319.0      Max.      :27.00      Max.      :1029.0
##      Gender      Marital_status      Car_loan      Personal_loan
## Length:1000      Length:1000      Min.      :0.000      Min.      :0.000
## Class :character      Class :character      1st Qu.:0.000      1st Qu.:0.000
## Mode  :character      Mode  :character      Median :0.000      Median :0.000
##                                     Mean      :0.353      Mean      :0.474
##                                     3rd Qu.:1.000      3rd Qu.:1.000
##                                     Max.      :1.000      Max.      :1.000
##      Home_loan      Education_loan      Emp_status      Amount
## Min.      :0.000      Min.      :0.000      Length:1000      Min.      : 244
## 1st Qu.:0.000      1st Qu.:0.000      Class :character      1st Qu.:1016
## Median :0.000      Median :0.000      Mode  :character      Median :1226
## Mean      :0.056      Mean      :0.112                                     Mean      :1219
## 3rd Qu.:0.000      3rd Qu.:0.000                                     3rd Qu.:1420
## Max.      :1.000      Max.      :1.000                                     Max.      :2362
##      Saving_amount      Emp_duration      Age      No_of_credit_acc
## Min.      :2082      Min.      : 0.00      Min.      :18.00      Min.      :1.000
## 1st Qu.:2951      1st Qu.: 15.00      1st Qu.:29.00      1st Qu.:1.000
## Median :3203      Median : 41.00      Median :32.00      Median :2.000
## Mean      :3179      Mean      : 49.39      Mean      :31.21      Mean      :2.546
## 3rd Qu.:3402      3rd Qu.: 85.00      3rd Qu.:34.00      3rd Qu.:3.000
## Max.      :4108      Max.      :120.00      Max.      :42.00      Max.      :9.000
```

Univariate Analysis

Univariate analysis examines each variable individually to understand its distribution, central tendency, and variability. This helps us identify patterns, spot anomalies, and get an initial feel for each variable before moving on to more complex analyses.

Categorical Variables

In this section, we conduct univariate analysis on the categorical variables in our dataset. We start by creating frequency tables for each categorical variable to see the count of each category, allowing us to quickly spot any dominant categories or imbalances. For example, we see that 30% of individuals have defaulted, and the majority of individuals do not have a home loan.

We then use bar plots to visually represent these counts for each variable. Through these plots, we can easily see distributions and gain insights, such as a higher representation of males compared to females, and a higher count of “single” individuals among the marital statuses. This helps us understand the dataset’s categorical structure before further analysis.

```
table(dataset$Default)

##
##    0    1
## 700 300

table(dataset$Car_loan)

##
##    0    1
## 647 353

table(dataset$Home_loan)

##
##    0    1
## 944   56

table(dataset$Personal_loan)

##
##    0    1
## 526 474

table(dataset$Education_loan)

##
##    0    1
## 888 112

table(dataset$Gender)

##
## Female   Male
##   310   690

table(dataset$Marital_status)
```

```
##
## Married   Single
##      548     452

table(dataset$Emp_status)

##
##   employed unemployed
##      308         692

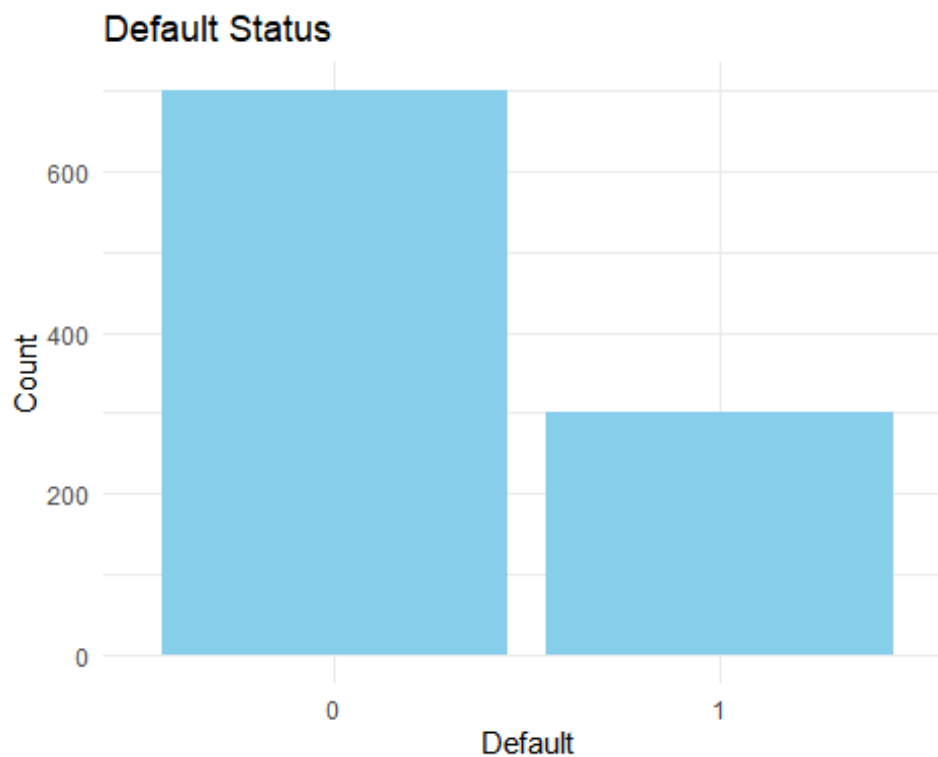
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

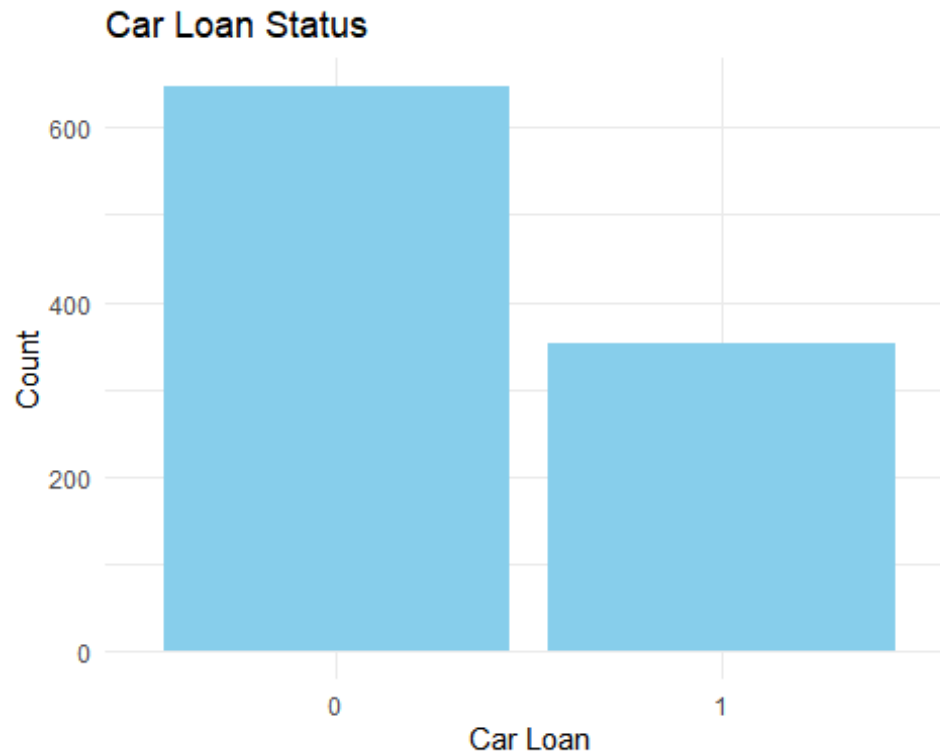
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

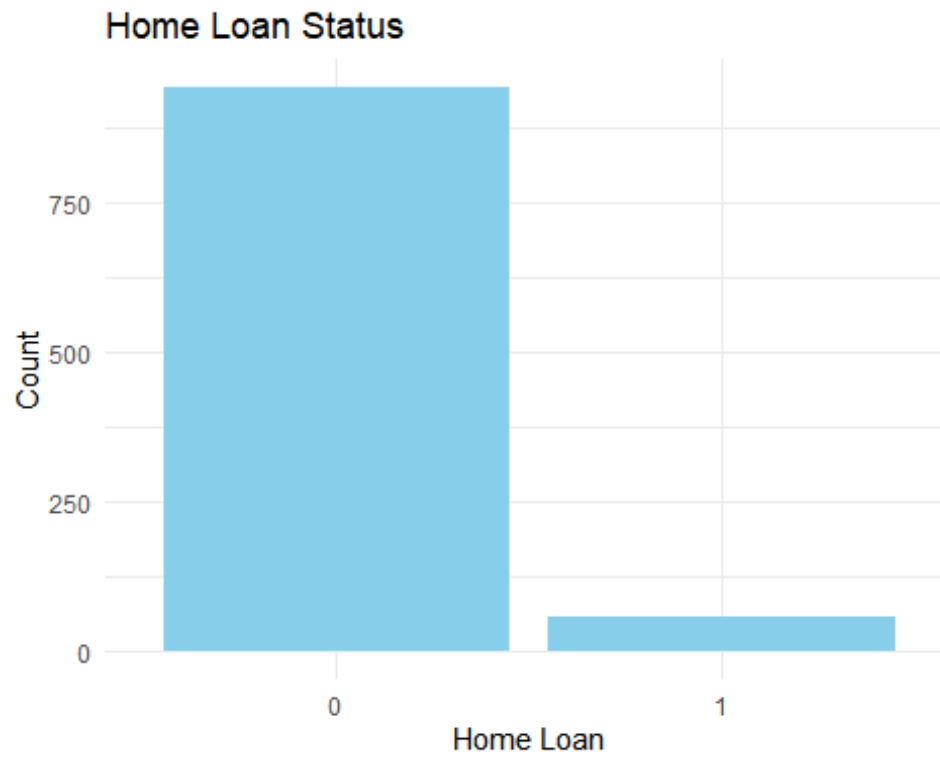
ggplot(dataset, aes(x = factor(Default))) +
  geom_bar(fill = "skyblue") +
  labs(title = "Default Status", x = "Default", y = "Count") +
  theme_minimal()
```



```
ggplot(dataset, aes(x = factor(Car_loan))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Car Loan Status", x = "Car Loan", y = "Count") +  
  theme_minimal()
```



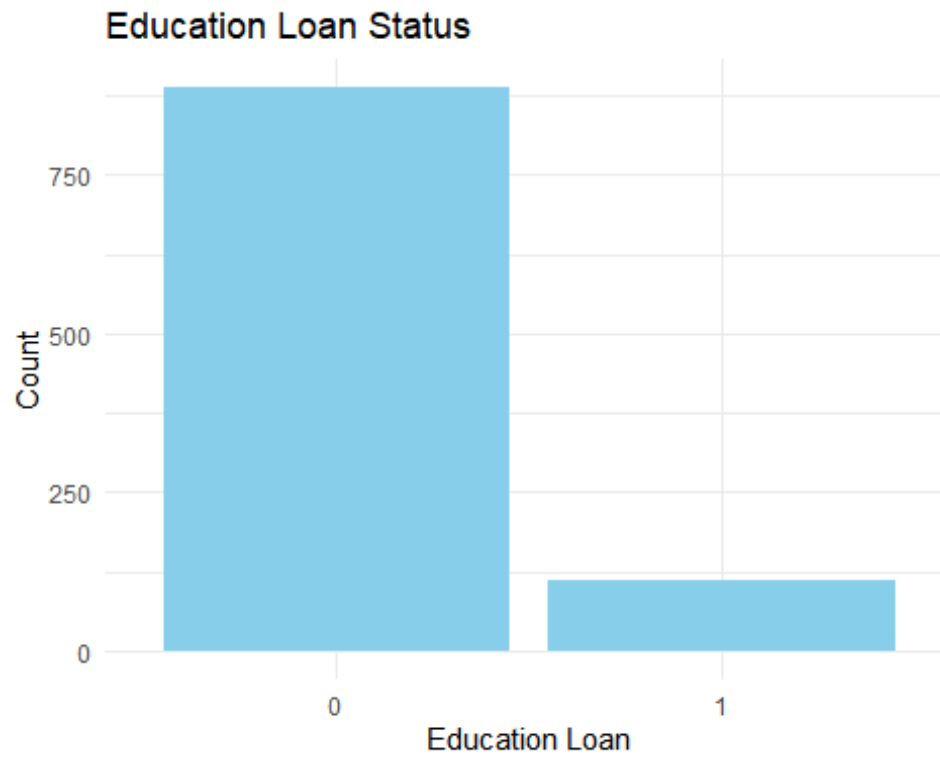
```
ggplot(dataset, aes(x = factor(Home_loan))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Home Loan Status", x = "Home Loan", y = "Count") +  
  theme_minimal()
```



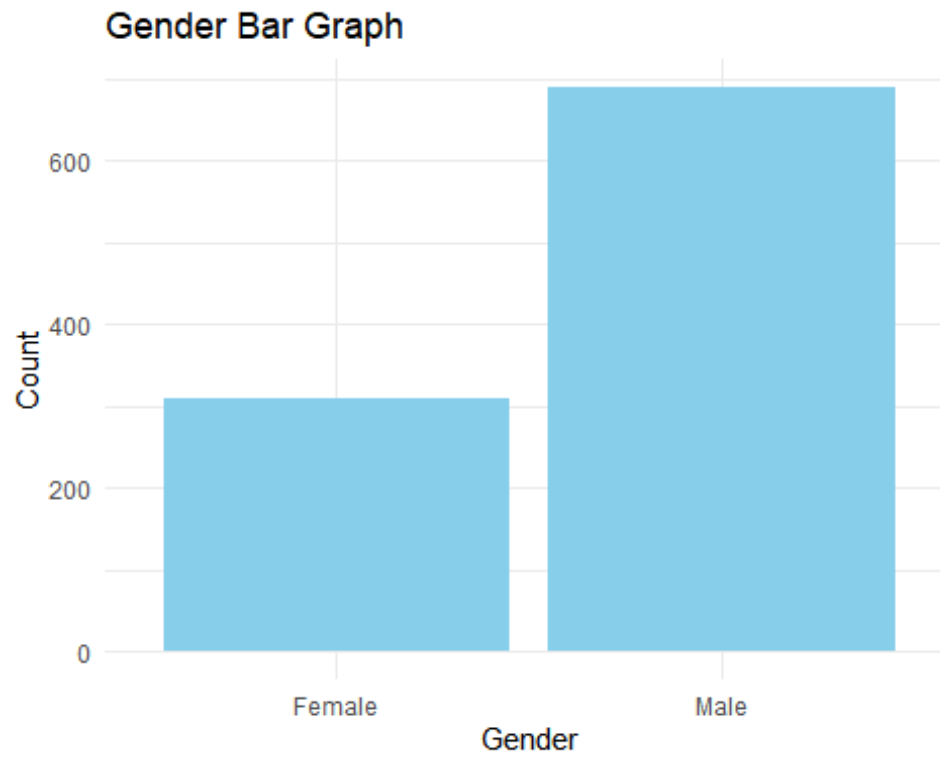
```
ggplot(dataset, aes(x = factor(Personal_loan))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Personal Loan Status", x = "Personal Loan", y = "Count") +  
  theme_minimal()
```



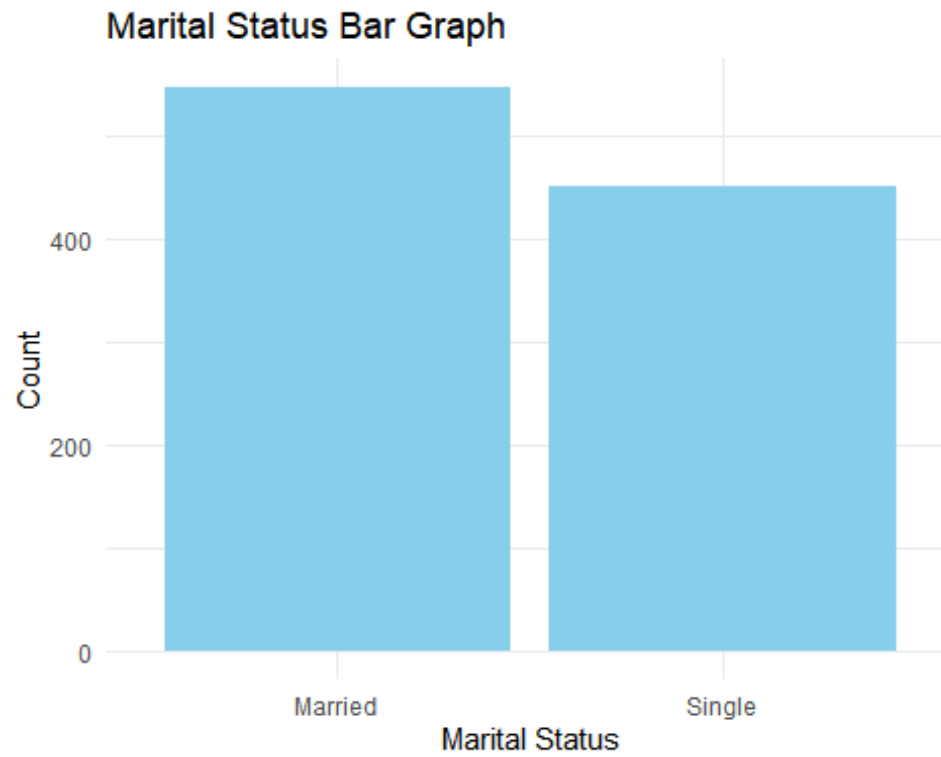
```
ggplot(dataset, aes(x = factor(Education_loan))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Education Loan Status", x = "Education Loan", y = "Count") +  
  theme_minimal()
```



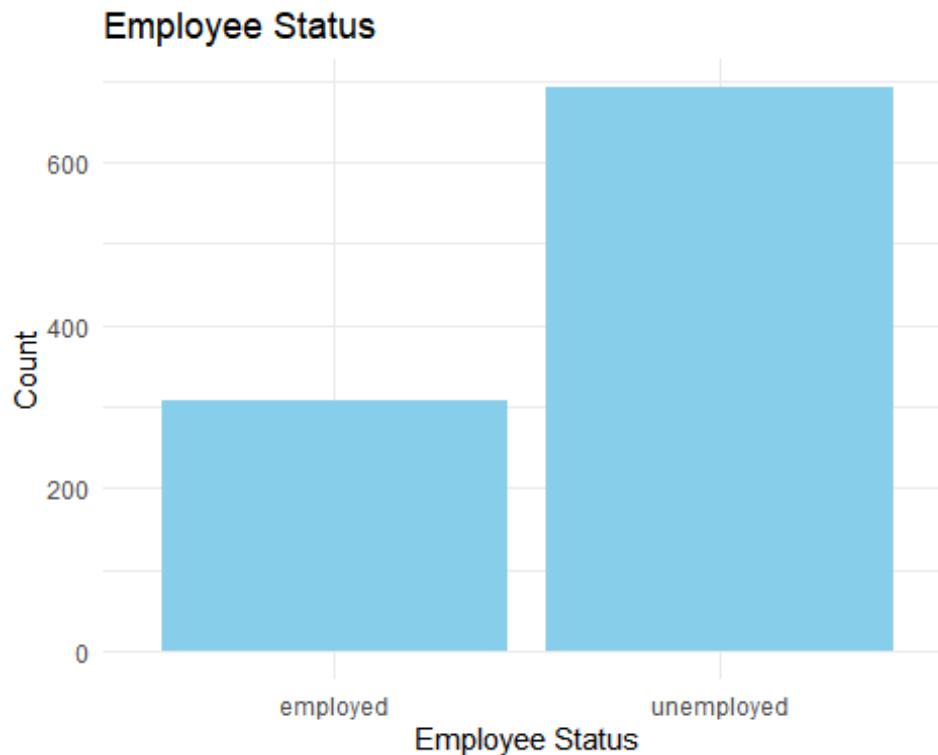
```
ggplot(dataset, aes(x = factor(Gender))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Gender Bar Graph", x = "Gender", y = "Count") +  
  theme_minimal()
```



```
ggplot(dataset, aes(x = factor(Marital_status))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Marital Status Bar Graph", x = "Marital Status", y = "Count")  
+  
  theme_minimal()
```



```
ggplot(dataset, aes(x = factor(Emp_status))) +  
  geom_bar(fill = "skyblue") +  
  labs(title = "Employee Status", x = "Employee Status", y = "Count") +  
  theme_minimal()
```



Numerical Variables

Checking Amount

The analysis of the `Checking_amount` variable indicates a generally normal distribution, as evidenced by the Shapiro-Wilk test. Summary statistics reveal a minimum value of -665.0, indicating overdrafts, while the first quartile (Q1) is 164.8, the median is 351.5, and the mean is slightly higher at 362.4, suggesting a right skew in the data. The interquartile range (IQR) is 388.75, with a standard deviation of approximately 300.90, indicating moderate spread and significant variability.

Outlier detection showed lower and upper bounds of -418.375 and 1136.625, respectively. Several negative values and high amounts fall outside this range, highlighting concerning overdrafts. A boxplot further confirms these findings, visually representing the IQR, median, and outliers.

```
shapiro.test(dataset$Checking_amount)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  dataset$Checking_amount  
## W = 0.99762, p-value = 0.1582
```

```
summary(dataset$Checking_amount)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -665.0   164.8   351.5   362.4   553.5  1319.0

IQR(dataset$Checking_amount, na.rm = TRUE)

## [1] 388.75

sd(dataset$Checking_amount, na.rm = TRUE)

## [1] 300.9012

lower_bound <- quantile(dataset$Checking_amount, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Checking_amount, na.rm = TRUE)
upper_bound <- quantile(dataset$Checking_amount, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Checking_amount, na.rm = TRUE)
lower_bound

##      25%
## -418.375

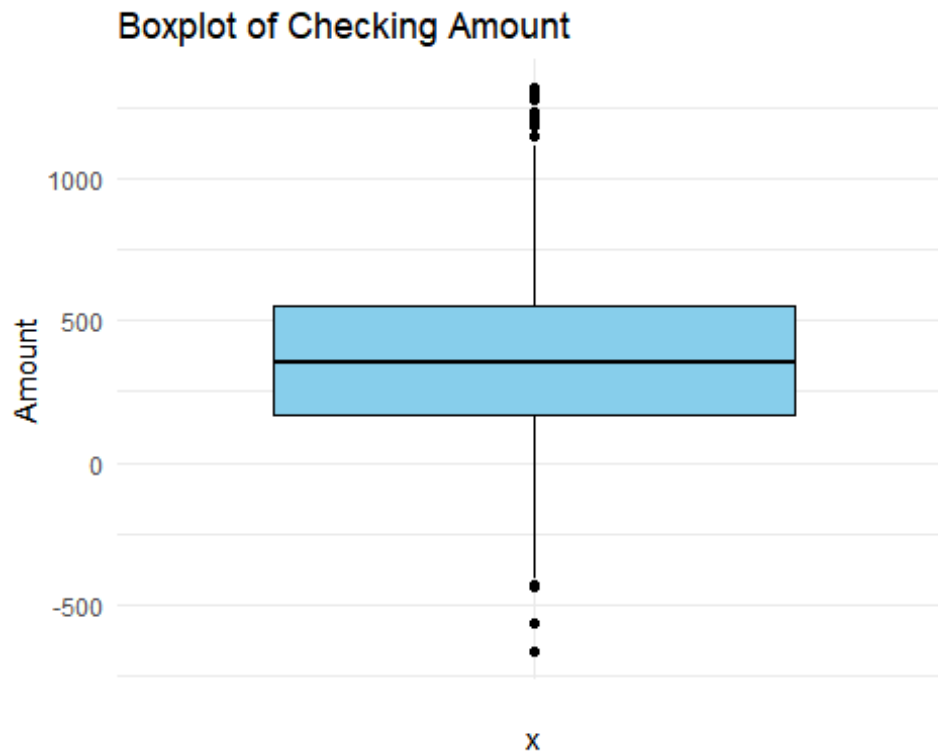
upper_bound

##      75%
## 1136.625

dataset[dataset$Checking_amount > upper_bound | dataset$Checking_amount <
lower_bound, "Checking_amount"]

## [1] 1151 1184 1213 1199 1275 1217 1296 -436 1319 1237 -567 -431 -665

ggplot(dataset, aes(x = "", y = Checking_amount)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of Checking Amount", y = "Amount") +
  theme_minimal()
```



Term

The Term variable reveals significant deviations from normality, as indicated by the Shapiro-Wilk test. Summary statistics show a range from 9.00 to 27.00, with a first quartile of 16.00, a median of 18.00, and a mean of 17.82, suggesting a slight right skew.

The interquartile range (IQR) is 4, with a standard deviation of approximately 3.24, reflecting moderate variability. Outlier detection identified bounds of 10 and 26, with values like 27 classified as outliers. A boxplot highlights the IQR and outliers, confirming the presence of extreme values that warrant further investigation.

```
shapiro.test(dataset$Term)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  dataset$Term  
## W = 0.99066, p-value = 5.593e-06
```

```
summary(dataset$Term)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##     9.00  16.00   18.00   17.82  20.00   27.00
```

```
IQR(dataset$Term, na.rm = TRUE)
```

```
## [1] 4
```

```

sd(dataset$Term, na.rm = TRUE)

## [1] 3.240567

lower_bound <- quantile(dataset$Term, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Term, na.rm = TRUE)
upper_bound <- quantile(dataset$Term, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Term, na.rm = TRUE)
lower_bound

## 25%
## 10

upper_bound

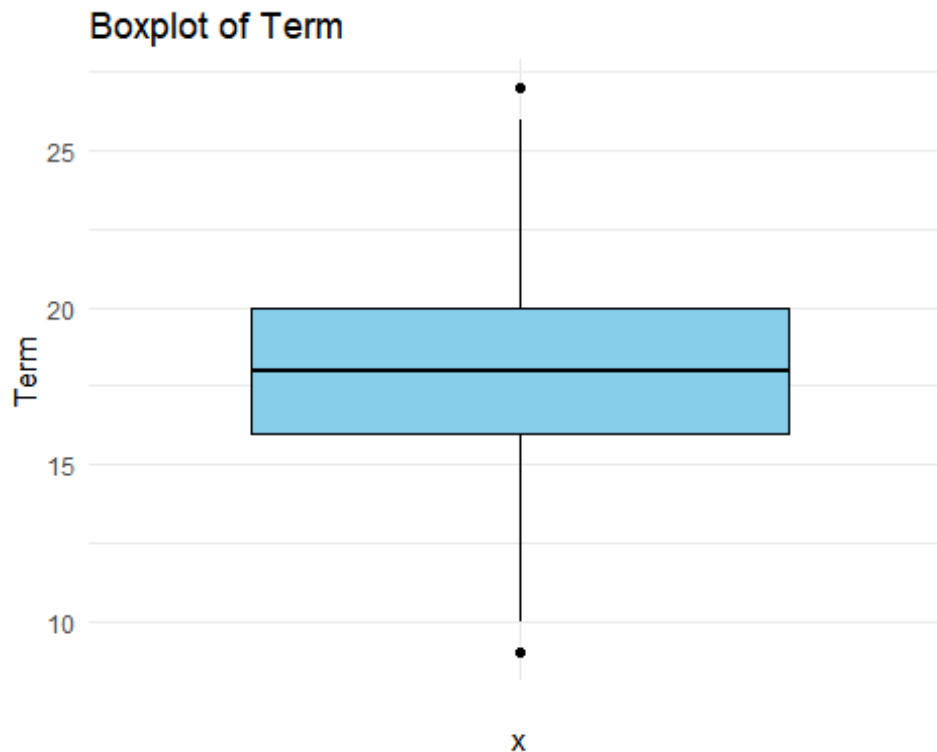
## 75%
## 26

dataset[dataset$Term > upper_bound | dataset$Term < lower_bound, "Term"]

## [1] 27 27 9 27 9 9

ggplot(dataset, aes(x = "", y = Term)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of Term", y = "Term") +
  theme_minimal()

```



Credit Score

The analysis of the Credit_score variable indicates a significant deviation from normality, like the Term variable, as evidenced by the Shapiro-Wilk test results. Summary statistics suggest a distribution with a slight right skew, given that the mean is higher than the median.

The interquartile range (IQR) and standard deviation indicate considerable variability in credit scores. Outlier detection identified several extreme values, which may impact the overall distribution. A histogram visualization further illustrates the frequency distribution of credit scores, confirming the presence of outliers and the general trend in the data.

```
shapiro.test(dataset$Credit_score)

##
##  Shapiro-Wilk normality test
##
## data:  dataset$Credit_score
## W = 0.95612, p-value < 2.2e-16

summary(dataset$Credit_score)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   376.0   725.8   770.5   760.5   812.0  1029.0

IQR(dataset$Credit_score, na.rm = TRUE)

## [1] 86.25

sd(dataset$Credit_score, na.rm = TRUE)

## [1] 77.55621

lower_bound <- quantile(dataset$Credit_score, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Credit_score, na.rm = TRUE)
upper_bound <- quantile(dataset$Credit_score, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Credit_score, na.rm = TRUE)
lower_bound

##      25%
## 596.375

upper_bound

##      75%
## 941.375

dataset[dataset$Credit_score > upper_bound | dataset$Credit_score <
lower_bound, "Credit_score"]

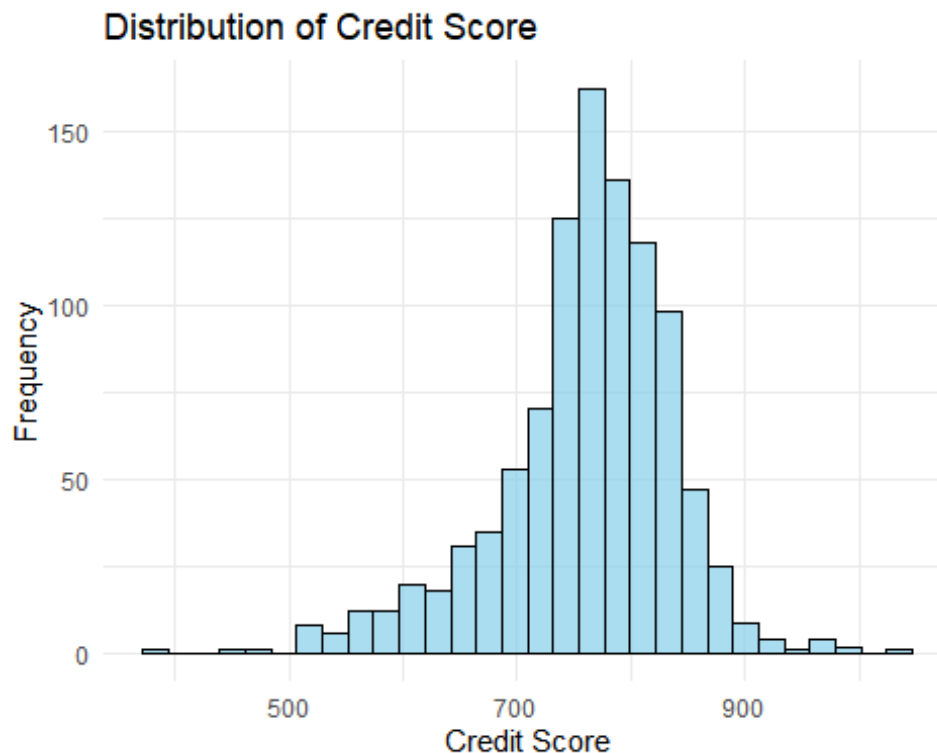
## [1] 563 531 528 1029 566 586 974 591 554 512 553 577 991 518
##      596
## [16] 534 528 526 559 540 469 579 565 586 590 566 561 569 532
```

```

575
## [31] 451 376 512 962 553 507 573 979 593 547 991 593 580 547
523
## [46] 964 582 562

ggplot(dataset, aes(x = Credit_score)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Credit Score", x = "Credit Score", y =
"Frequency") +
  theme_minimal()

```



Amount

The evaluation of the Amount variable indicates a distribution that shows nearly normal distribution, as evidenced by the Shapiro-Wilk test results, which do not suggest significant deviations. Summary statistics show a mild right skew, with the mean slightly exceeding the median.

The calculated interquartile range (IQR) and standard deviation reflect a moderate degree of variability in the dataset. Outlier analysis revealed several extreme values, which could influence the overall distribution pattern. A histogram visualization complements these findings, depicting the frequency distribution of amounts and underscoring the presence of outliers. Overall, while the Amount variable generally follows a normal distribution, investigating the identified outliers could provide further insights into their effects on the analysis.

```
shapiro.test(dataset$Amount)
```

```
##
## Shapiro-Wilk normality test
##
## data: dataset$Amount
## W = 0.99879, p-value = 0.7495

summary(dataset$Amount)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      244   1016   1226   1219   1420   2362

IQR(dataset$Amount, na.rm = TRUE)

## [1] 403.75

sd(dataset$Amount, na.rm = TRUE)

## [1] 305.7537

lower_bound <- quantile(dataset$Amount, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Amount, na.rm = TRUE)
upper_bound <- quantile(dataset$Amount, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Amount, na.rm = TRUE)
lower_bound

##      25%
## 410.375

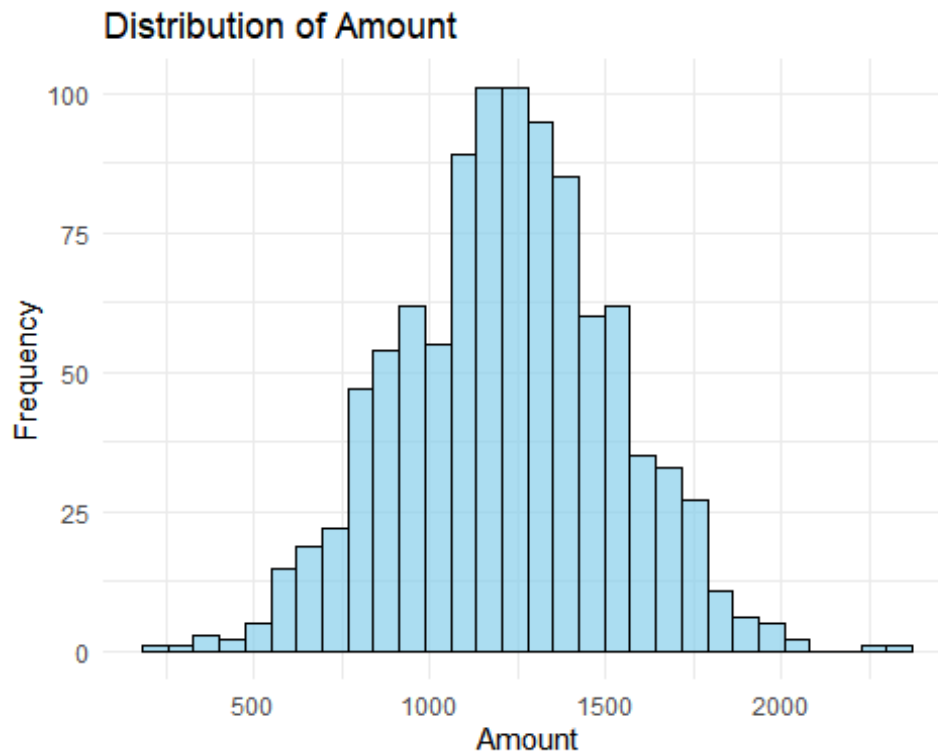
upper_bound

##      75%
## 2025.375

dataset[dataset$Amount > upper_bound | dataset$Amount < lower_bound,
"Amount"]

## [1] 2052 2066 395 244 316 405 385 386 2362 2259

ggplot(dataset, aes(x = Amount)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Amount", x = "Amount", y = "Frequency") +
  theme_minimal()
```



Saving Amount

The Saving_amount variable appears nearly normal, according to the Shapiro-Wilk test. Summary statistics indicate a slight right skew, with the mean exceeding the median.

Moderate variability is evident from the interquartile range (IQR) and standard deviation. Outlier detection revealed a few extreme values that could affect the distribution. A histogram highlights these outliers, suggesting that further investigation may provide additional insights into the data.

```
shapiro.test(dataset$Saving_amount)

##
##  Shapiro-Wilk normality test
##
## data:  dataset$Saving_amount
## W = 0.99716, p-value = 0.07442

summary(dataset$Saving_amount)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2082  2951    3203    3179  3402    4108

IQR(dataset$Saving_amount, na.rm = TRUE)

## [1] 451.25

sd(dataset$Saving_amount, na.rm = TRUE)
```

```
## [1] 339.5498

lower_bound <- quantile(dataset$Saving_amount, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Saving_amount, na.rm = TRUE)
upper_bound <- quantile(dataset$Saving_amount, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Saving_amount, na.rm = TRUE)
lower_bound

##      25%
## 2274.125

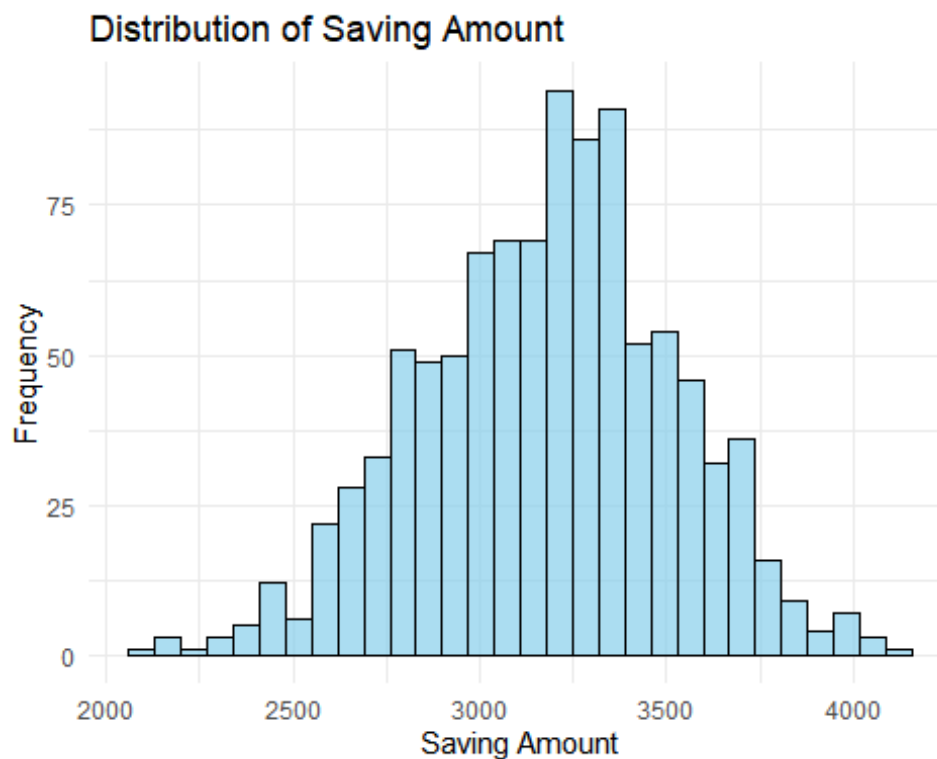
upper_bound

##      75%
## 4079.125

dataset[dataset$Saving_amount > upper_bound | dataset$Saving_amount <
lower_bound, "Saving_amount"]

## [1] 2191 4108 2191 2082 2145 2274 2248

ggplot(dataset, aes(x = Saving_amount)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Saving Amount", x = "Saving Amount", y =
"Frequency") +
  theme_minimal()
```



Employee Duration

The Emp_duration variable displays a non-normal distribution, as evidenced by the Shapiro-Wilk test. The median is lower than the mean, indicating a right skew. There is substantial variability, with outliers identified that may influence the distribution. A boxplot further illustrates these characteristics, highlighting the need for further examination of the outliers and their impact.

```
shapiro.test(dataset$Emp_duration)

##
##  Shapiro-Wilk normality test
##
## data:  dataset$Emp_duration
## W = 0.91576, p-value < 2.2e-16

summary(dataset$Emp_duration)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00  15.00   41.00   49.39   85.00  120.00

IQR(dataset$Emp_duration, na.rm = TRUE)

## [1] 70

sd(dataset$Emp_duration, na.rm = TRUE)

## [1] 37.75952

lower_bound <- quantile(dataset$Emp_duration, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$Emp_duration, na.rm = TRUE)
upper_bound <- quantile(dataset$Emp_duration, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$Emp_duration, na.rm = TRUE)
lower_bound

## 25%
## -90

upper_bound

## 75%
## 190

dataset[dataset$Emp_duration > upper_bound | dataset$Emp_duration <
lower_bound, "Emp_duration"]

## numeric(0)

ggplot(dataset, aes(x = "", y = Emp_duration)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of Employment Duration", y = "Employment Duration") +
  theme_minimal()
```



Number of Credit Accounts

The `No_of_credit_acc` variable shows a non-normal distribution, as indicated by the Shapiro-Wilk test results. The median is lower than the mean, suggesting a right skew. The range of credit accounts is narrow, with a notable number of outliers present, as revealed by the boxplot. This variability suggests that while most individuals have few credit accounts, some have significantly more.

```
shapiro.test(dataset$No_of_credit_acc)

##
##  Shapiro-Wilk normality test
##
## data:  dataset$No_of_credit_acc
## W = 0.81855, p-value < 2.2e-16

summary(dataset$No_of_credit_acc)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000  1.000   2.000   2.546  3.000   9.000

IQR(dataset$No_of_credit_acc, na.rm = TRUE)

## [1] 2

sd(dataset$No_of_credit_acc, na.rm = TRUE)

## [1] 1.652457
```

```

lower_bound <- quantile(dataset$No_of_credit_acc, 0.25, na.rm = TRUE) - 1.5 *
IQR(dataset$No_of_credit_acc, na.rm = TRUE)
upper_bound <- quantile(dataset$No_of_credit_acc, 0.75, na.rm = TRUE) + 1.5 *
IQR(dataset$No_of_credit_acc, na.rm = TRUE)
lower_bound

## 25%
## -2

upper_bound

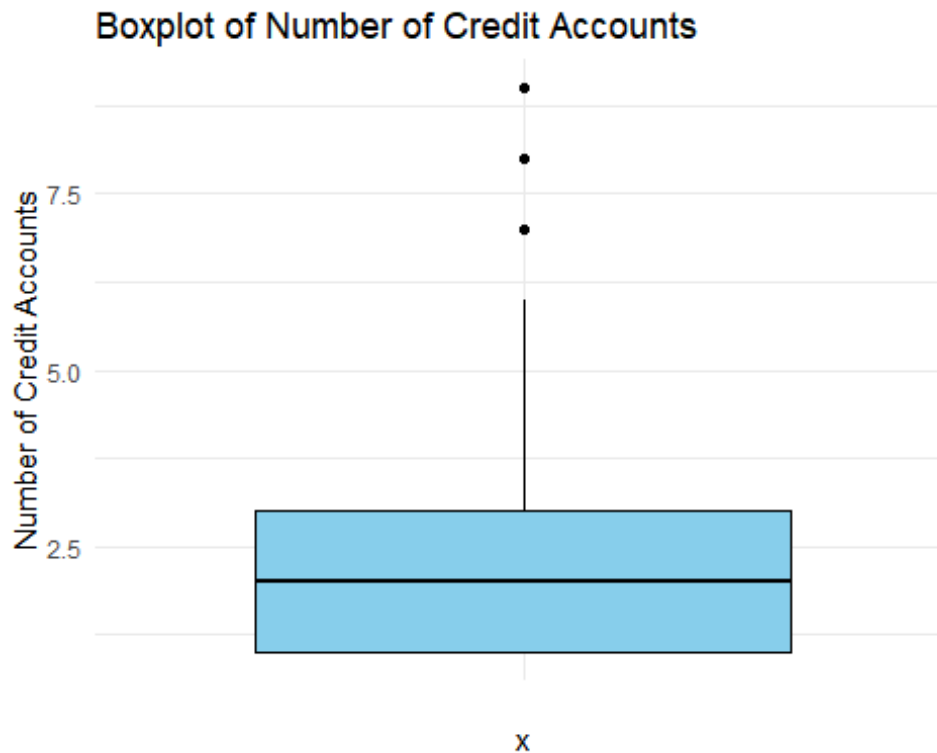
## 75%
## 6

dataset[dataset$No_of_credit_acc > upper_bound | dataset$No_of_credit_acc <
lower_bound, "No_of_credit_acc"]

## [1] 9 9 9 9 7 8 8 7 9 9 7 8 7 9 9 9 8 7 8 7 7 9 9 9 9 7 9 8

ggplot(dataset, aes(x = "", y = No_of_credit_acc)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of Number of Credit Accounts", y = "Number of Credit
Accounts") +
  theme_minimal()

```



Bivariate Analysis

Amount x Employment Status

The analysis for Amount by Emp_status uses an ANOVA to test if the average loan amount differs significantly across employment statuses.

The results indicate that the average loan amounts for both employed and unemployed individuals are nearly identical, with means around 1217 for employed and 1219 for unemployed. The standard deviations are also similar, suggesting consistent variability in loan amounts within each group.

The ANOVA results show a F-value of 0.014 with a p-value of 0.905, indicating no statistically significant difference in average loan amounts based on employment status. This suggests that employment status does not significantly affect the amount of loans granted. The accompanying boxplot visually reinforces this finding, demonstrating overlapping distributions for both employment categories, confirming that loan amounts are largely similar regardless of employment status.

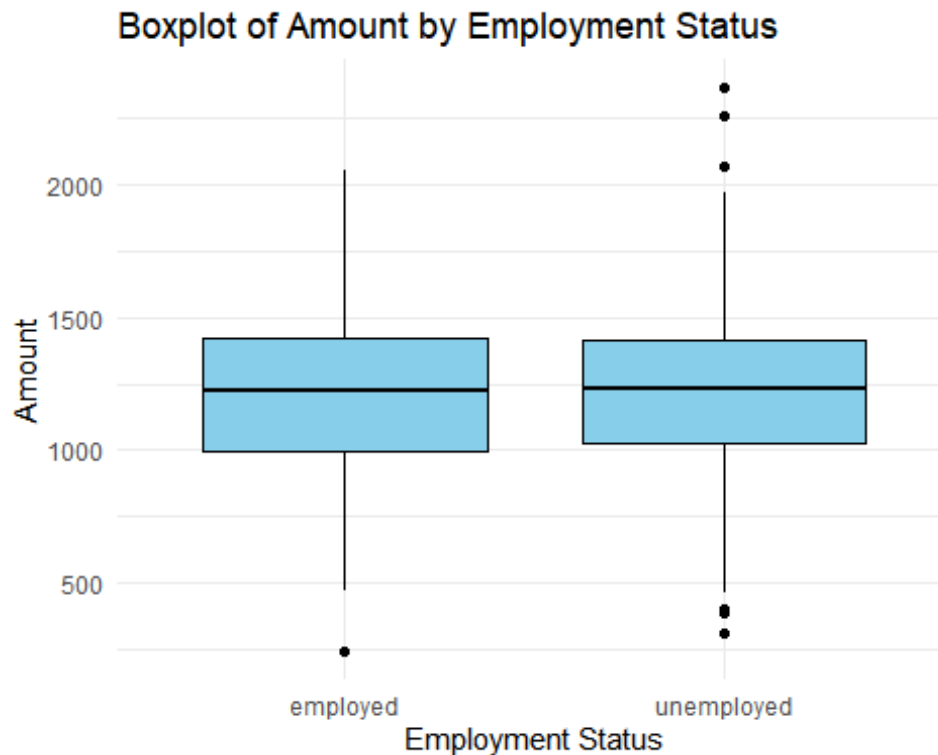
```
dataset %>% group_by(Emp_status) %>% summarise(mean_amount = mean(Amount,
na.rm = TRUE), sd_amount = sd(Amount, na.rm = TRUE))

## # A tibble: 2 × 3
##   Emp_status mean_amount sd_amount
##   <chr>          <dbl>      <dbl>
## 1 employed      1217.        305.
## 2 unemployed    1219.        306.

#Anova
aov_result <- aov(Amount ~ Emp_status, data = dataset)
summary(aov_result)

##              Df    Sum Sq Mean Sq F value Pr(>F)
## Emp_status    1      1337    1337  0.014  0.905
## Residuals   998 93390503    93578

ggplot(dataset, aes(x = Emp_status, y = Amount)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of Amount by Employment Status", x = "Employment
Status", y = "Amount") +
  theme_minimal()
```



Credit Score x Savings Amount

For Credit_score and Saving_amount, a linear regression model examines the relationship between these two variables. The correlation coefficient provides an initial assessment of the strength and direction of their association.

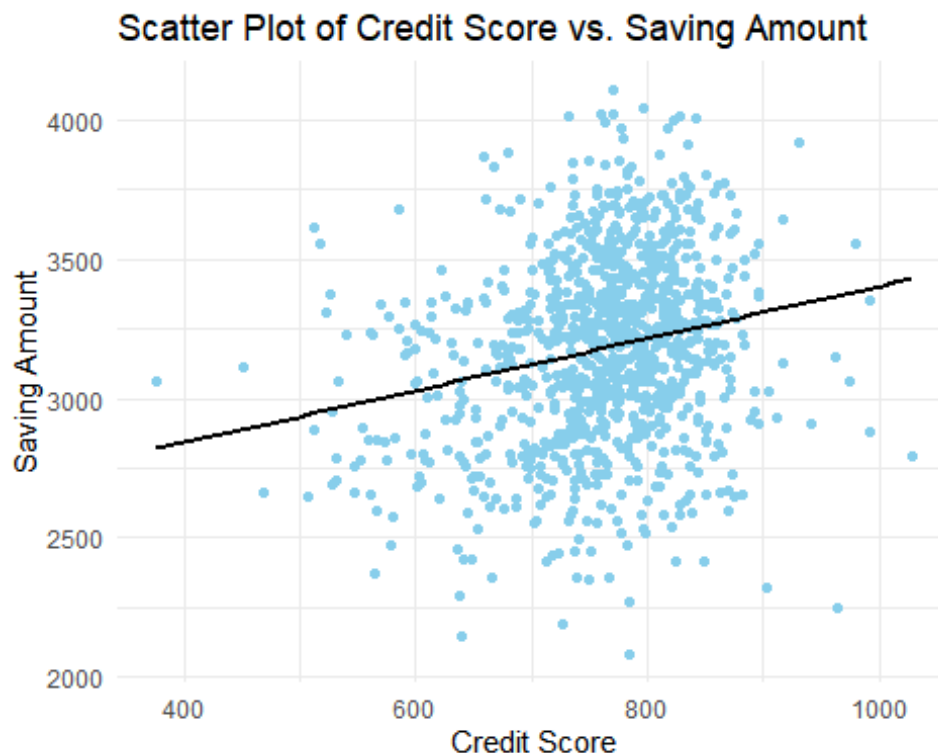
The analysis reveals a weak positive correlation between credit score and savings amount. The linear regression indicates that an increase in credit score is associated with a rise in savings amount, with the model showing statistical significance. The scatter plot illustrates a positive trend, although the data points are widely dispersed around the regression line, indicating variability in the relationship.

```
correlation <- cor(dataset$Credit_score, dataset$Saving_amount, use =  
"complete.obs")  
correlation  
## [1] 0.2138242  
  
#Linear Regression  
lm_result <- lm(Saving_amount ~ Credit_score, data = dataset)  
summary(lm_result)  
##  
## Call:  
## lm(formula = Saving_amount ~ Credit_score, data = dataset)  
##  
## Residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -1121.80 -224.49   13.49   227.29   919.82
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2467.3503    103.4877   23.842 < 2e-16 ***
## Credit_score    0.9361     0.1354    6.915 8.35e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 331.9 on 998 degrees of freedom
## Multiple R-squared:  0.04572,    Adjusted R-squared:  0.04476
## F-statistic: 47.82 on 1 and 998 DF,  p-value: 8.349e-12

ggplot(dataset, aes(x = Credit_score, y = Saving_amount)) +
  geom_point(color = "skyblue") +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(title = "Scatter Plot of Credit Score vs. Saving Amount", x = "Credit
Score", y = "Saving Amount") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



Terms x Age

The correlation analysis indicates a moderate negative relationship between loan term and age, suggesting that as the loan term increases, age tends to decrease. The linear regression

model further supports this finding, revealing that loan term significantly predicts age, with a notable decrease in age for each additional term unit. The scatter plot visually represents this trend, displaying a downward slope in the data points, although there is considerable variability around the fitted line.

```
correlation_terms_age <- cor(dataset$Term, dataset$Age, use = "complete.obs")
print(correlation_terms_age)

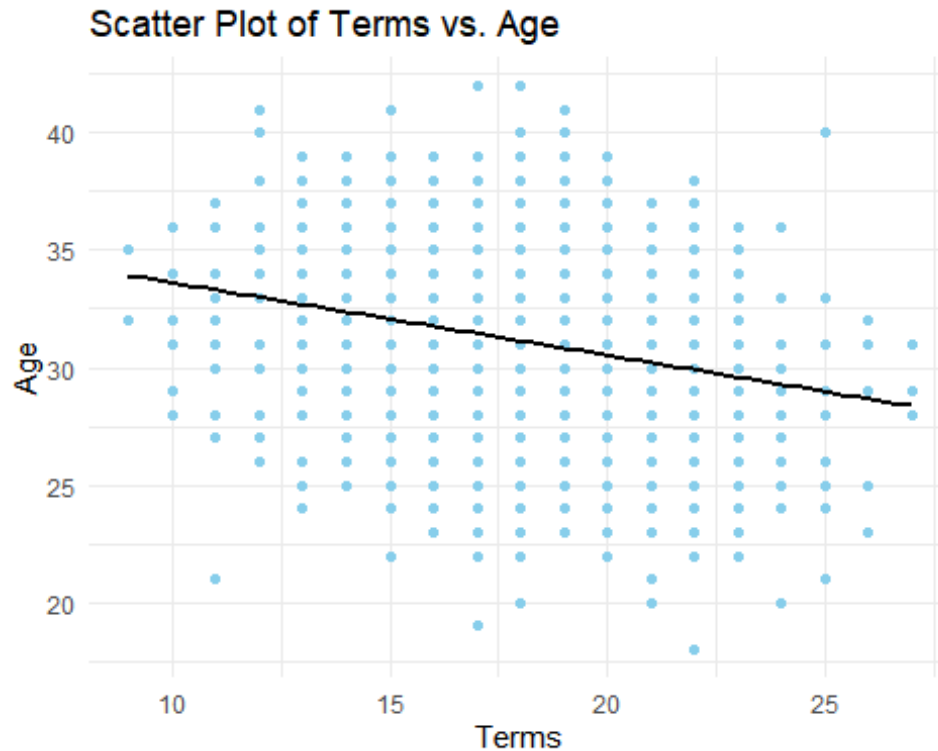
## [1] -0.2443853

linear_model <- lm(Age ~ Term, data = dataset)
summary(linear_model)

##
## Call:
## lm(formula = Age ~ Term, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.4606  -2.6085   0.2307   2.7742  11.0089
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  36.70821    0.70202  52.289  < 2e-16 ***
## Term        -0.30868    0.03877  -7.962 4.59e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.971 on 998 degrees of freedom
## Multiple R-squared:  0.05972,    Adjusted R-squared:  0.05878
## F-statistic: 63.39 on 1 and 998 DF,  p-value: 4.592e-15

ggplot(dataset, aes(x = Term, y = Age)) +
  geom_point(color = "skyblue") +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(title = "Scatter Plot of Terms vs. Age", x = "Terms", y = "Age") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



Credit Score x Amount

The correlation analysis shows a very weak negative correlation between credit score and loan amount, indicating that as credit scores increase, loan amounts tend to decrease slightly. This suggests that individuals with higher credit scores may not necessarily be associated with larger loan amounts, which could imply that other factors are influencing loan size.

Further exploration through linear regression reveals that credit score is a statistically significant predictor of loan amount, but the relationship is minimal. The regression model indicates that changes in credit score correspond to minor fluctuations in the loan amount. The results are supported by the scatter plot, where the fitted line illustrates a slight downward trend in the relationship, reinforcing the idea that while credit scores do have some impact on loan amounts.

```
correlation_credit_amount <- cor(dataset$Credit_score, dataset$Amount, use =
"complete.obs")
correlation_credit_amount

## [1] -0.07839842

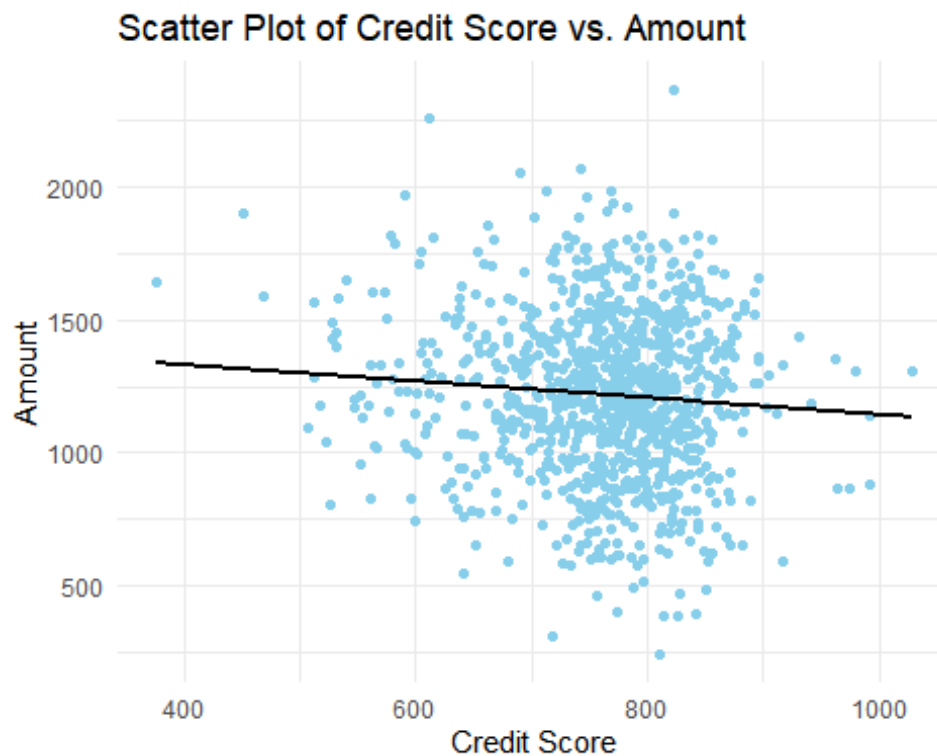
lm_credit_amount <- lm(Amount ~ Credit_score, data = dataset)
summary(lm_credit_amount)

##
## Call:
## lm(formula = Amount ~ Credit_score, data = dataset)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -959.37 -202.90    4.98   202.33 1162.64
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1453.7240    95.1000   15.286  <2e-16 ***
## Credit_score   -0.3091     0.1244   -2.484   0.0131 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 305 on 998 degrees of freedom
## Multiple R-squared:  0.006146,    Adjusted R-squared:  0.00515
## F-statistic: 6.172 on 1 and 998 DF,  p-value: 0.01314

ggplot(dataset, aes(x = Credit_score, y = Amount)) +
  geom_point(color = "skyblue") +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(title = "Scatter Plot of Credit Score vs. Amount", x = "Credit Score",
y = "Amount") +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



Default x Marital Status

The analysis explores the relationship between marital status and default rates using a contingency table and Pearson's Chi-squared test. The contingency table displays the counts of individuals who defaulted and those who did not, categorized by marital status—Married and Single. The Chi-squared test results indicate a statistically significant association between marital status and default rates. Specifically, married individuals appear to have a lower likelihood of defaulting compared to singles. This finding highlights the potential influence of marital status on financial behavior, suggesting that marital dynamics could play a role in default risk. The accompanying bar chart visually represents the differences in default counts between the two groups, further illustrating the results of the analysis.

```
dataset$Marital_status <- as.factor(dataset$Marital_status)
dataset$Default <- as.factor(dataset$Default)

contingency_table <- table(dataset$Marital_status, dataset$Default)
contingency_table

##
##           0    1
## Married 402 146
## Single  298 154

chi_squared_test <- chisq.test(contingency_table)
chi_squared_test

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 6.1598, df = 1, p-value = 0.01307

ggplot(dataset, aes(x = Marital_status, fill = Default)) +
  geom_bar(position = "dodge") +
  labs(title = "Default Count by Marital Status", x = "Marital Status", y =
"Count") +
  theme_minimal()
```

